Introduction to Network Administration and Physical Network Development

Overview of Network Administration

Network Administration refers to the process of managing, maintaining, monitoring, and securing computer networks to ensure efficient communication, availability of resources, and reliable system performance.

It involves managing both hardware and software components of a network including servers, routers, switches, operating systems, and security systems.

Objectives of Network Administration

The main objectives include:

1. **Ensuring Network Availability**
    - Networks should be operational with minimal downtime.
2. **Maintaining Network Performance**
    - Ensuring efficient data transmission and system response time.
3. **Managing Network Resources**
    - Allocating shared resources such as printers, storage, and internet access.
4. **Ensuring Security**
    - Protecting network resources from unauthorized access, malware, and cyber threats.
5. **Monitoring Network Activity**
    - Detecting problems, performance issues, and security breaches.

Roles of a Network Administrator

A **network administrator** is responsible for the day-to-day management of the network infrastructure.

Typical responsibilities include:

- Installing and configuring network hardware
- Managing network servers
- Monitoring network traffic
- Troubleshooting connectivity problems

- Implementing security measures
- Managing user accounts and permissions
- Performing network backups and updates

Key Functions of Network Administration

1. Network Configuration

Setting up network devices such as routers, switches, and servers.

2. Network Monitoring

Tracking network performance and identifying faults.

3. Network Maintenance

Updating software, replacing faulty devices, and maintaining system reliability.

4. Network Security Management

Protecting network infrastructure through firewalls, encryption, and access controls.

Developing the Physical Network

Physical network development refers to the process of designing and installing the hardware infrastructure of a computer network.

It involves planning how network devices will be connected and how data will physically move between systems.

Components of a Physical Network

1. Network Devices

Examples include:

- Routers
- Switches
- Hubs
- Access points
- Firewalls

2. Transmission Media

The physical medium through which data travels.

Examples:

- Twisted pair cables
- Fiber optic cables
- Coaxial cables
- Wireless signals

3. Network Interface Cards (NICs)

Hardware components that allow computers to connect to a network.

4. Servers

Computers that provide services such as:

- File storage
- Email services
- Web hosting
- Database management

Steps in Developing a Physical Network

1. Identify network requirements
2. Determine the network topology
3. Select networking hardware
4. Install cabling infrastructure
5. Configure network devices
6. Test the network connectivity

Preliminary Groundwork in Network Design

Before implementing a network, careful preparation must be done.

Needs Assessment

Understanding the organization's communication requirements.

Factors considered include:

- Number of users

- Type of applications used
- Data storage needs
- Security requirements
- Internet usage

Site Survey

A **site survey** involves examining the physical environment where the network will be installed.

Key considerations include:

- Building layout
- Distance between departments
- Electrical interference
- Cabling routes
- Equipment rooms

Budget Planning

Network design must consider financial constraints.

Costs include:

- Networking hardware
- Software licenses
- Installation
- Maintenance
- Training

Scalability Considerations

A network should be designed to **allow future expansion**.

This includes:

- Supporting more users
- Adding new devices
- Increasing storage capacity

Network Foundations and Concepts

What is a Computer Network?

A **computer network** is a collection of interconnected devices that communicate and share resources.

Examples of shared resources include:

- Files
- Printers
- Applications
- Internet access

Types of Networks

Local Area Network (LAN)

A network within a small geographic area such as:

- Offices
- Schools
- Homes

Metropolitan Area Network (MAN)

A network covering a city or large campus.

Wide Area Network (WAN)

A network that spans large geographic areas such as countries or continents.

The **Internet** is the largest example of a WAN.

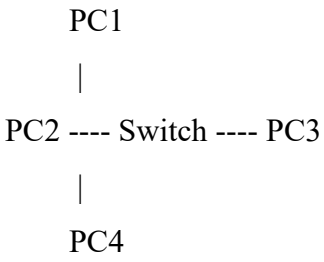
Network Topologies

A **network topology** refers to the arrangement of network devices.

Common topologies include:

1. Star Topology

Diagram



Explanation

In a **Star Topology**, all devices are connected to a **central device** such as a **switch or hub**. Communication between devices passes through this central device. When one computer sends data, it goes to the switch, which then forwards it to the correct destination.

This topology is commonly used in modern **Local Area Networks (LANs)** because it simplifies network management and troubleshooting.

Advantages

1. **Easy to Manage**
   The network is centrally controlled, making monitoring and maintenance easier.
2. **Easy Fault Detection**
   If one device fails, it does not affect the rest of the network.

Disadvantages

1. **Central Device Failure**
   If the hub or switch fails, the entire network stops working.
2. **Higher Installation Cost**
   Requires more cables compared to some other network topologies.

2. Bus Topology

Diagram

PC1 ---- PC2 ---- PC3 ---- PC4 ---- PC5

(Single Backbone Cable)

Explanation

In a **Bus Topology**, all devices are connected to a **single main communication cable** called the **backbone**. Data sent by one device travels along the cable and is received by all devices, but only the intended recipient processes the data.

Bus topology was widely used in early computer networks but is less common today due to reliability limitations.

Advantages

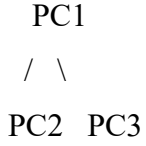
1. **Low Cost**
   Requires less cabling, making it cheaper to install.
2. **Simple Network Design**
   Easy to set up for small networks.

Disadvantages

1. **Single Cable Failure**
   If the backbone cable fails, the entire network stops working.
2. **Reduced Performance with More Devices**
   As more devices are added, network traffic increases, causing slower performance.

3. Ring Topology

Diagram



```
\   /
 PC4
```

Explanation

In a **Ring Topology**, devices are connected in a **circular structure**, forming a closed loop. Data travels around the ring from one device to another until it reaches the intended destination.

Each device acts as a **repeater**, receiving data and forwarding it to the next device in the ring.

Advantages

1. **Orderly Data Transmission**
   Data flows in one direction, reducing chances of data collisions.
2. **Better Performance than Bus in Some Cases**
   Network traffic is more organized.

Disadvantages

1. **Network Failure if One Device Fails**
   A single device failure can disrupt the entire network.
2. **Difficult to Troubleshoot**
   Locating faults in the ring can be complicated.

4. Mesh Topology

Diagram

```
   PC1
  / | \
PC2--|--PC3
  \ | /
   PC4
```

Explanation

In a **Mesh Topology**, each device is connected to **multiple other devices**, creating multiple communication paths. Data can travel through different routes to reach its destination.

This topology is used in networks where **high reliability and fault tolerance** are required.

Advantages

1. **High Reliability**
   If one link fails, data can travel through another path.
2. **Better Performance**
   Multiple paths allow efficient data transmission.

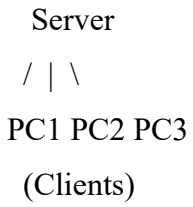
Disadvantages

1. **Expensive to Install**
   Requires a large number of cables and ports.
2. **Complex Management**
   Difficult to configure and maintain due to many connections.

Network Models

1. Client–Server Model

Diagram



Explanation

In the **Client–Server Model**, a **central server** provides services and resources to multiple client computers. Clients send requests to the server, and the server processes those requests and returns the results.

Examples of Server Services

- File servers
- Email servers
- Web servers

Advantages

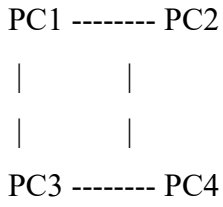
1. **Centralized Management**
   Administrators can manage users, security, and resources from the server.
2. **Improved Security**
   Data is centrally stored and controlled.

Disadvantages

1. **Higher Cost**
   Requires dedicated server hardware and specialized software.
2. **Server Failure Stops Services**
   If the server crashes, clients cannot access network services.

2. Peer-to-Peer Model

Diagram



Explanation

In a **Peer-to-Peer (P2P) Network**, all computers have equal roles. Each computer can share resources such as files, printers, or internet connections directly with others.

There is **no central server controlling the network**.

Advantages

1. **Low Cost**

   Does not require a dedicated server.
2. **Easy Setup**

   Simple to install and configure.

Disadvantages

1. **Weak Security**

   Lack of centralized control makes it harder to enforce security policies.
2. **Poor Performance for Large Networks**

   Becomes difficult to manage as the network grows.

Organizational Network Policies and Procedures

Network policies are rules and guidelines that control how an organization's network is used and managed.

These policies ensure **security, efficiency, and proper use of network resources**.

Types of Network Policies

1. Acceptable Use Policy (AUP)

Defines how employees are allowed to use the organization's network.

Examples:

- Internet usage rules
- Email usage guidelines
- Restrictions on downloading software

2. Security Policy

Specifies procedures to protect network resources.

Examples:

- Password requirements

- Data encryption rules
- Access control policies

3. Backup Policy

Defines how data backups are performed.

Includes:

- Backup frequency
- Storage location
- Recovery procedures

4. Access Control Policy

Defines who can access specific network resources.

Examples:

- User authentication
- Role-based access control
- Permission management

Importance of Network Policies

Network policies help organizations to:

- Protect sensitive data
- Prevent misuse of network resources
- Improve network management
- Ensure compliance with regulations
- Reduce cybersecurity risks